

$$35. E(X|\eta=\frac{n}{2}) = \frac{n+1}{2} \quad \text{而 } n \sim U[0, a]$$

$$\therefore E(X|\eta) \sim U[\frac{a}{2}, a]$$

$$41. E(X_1 | X_1 < X_2) = \frac{E(X_1 I_{X_1 < X_2})}{P(X_1 < X_2)}$$

$$= \frac{\int_0^1 \int_0^y x f(x, y) dx dy}{\int_0^1 \int_0^y f(x, y) dx dy}$$

由于 ~~独立~~ X_1, X_2 独立, $\therefore f(x, y) = f_{X_1}(x) f_{X_2}(y) = \lambda_1 \lambda_2 e^{-\lambda_1 x - \lambda_2 y}$

$$\therefore \text{上式} = \frac{\frac{1}{(\lambda_1 + \lambda_2)^2}}{\frac{1}{\lambda_1 + \lambda_2}} = \frac{1}{\lambda_1 + \lambda_2}$$

$$49. f_{X,Y}(x, y) = f_X(x) f_Y(y) = e^{-x-y} \quad (x > 0, y > 0)$$

$$f_{u,v}(u, v) = f_{X,Y}\left(\frac{uv}{v+1}, \frac{u}{v+1}\right) |J(\frac{uv}{v+1}, \frac{u}{v+1})|$$

$$= \begin{cases} ue^{-u} \frac{1}{(1+v)^2} & u > 0, v > 0 \\ 0 & \text{其他} \end{cases}$$

~~对~~ $f_{u,v}(u, v) = u e^{-u} \cdot \frac{1}{(1+v)^2} \quad \therefore u, v \text{ 独立}$

$$\begin{aligned}
 50. \quad f_{u,v}(u,v) &= f_{x,y}(u,v) |J_1(u,v)| + f_{x,y}(v,u) |J_1(u,v)| \\
 &= \begin{cases} \frac{12}{7} (u+v)^2 & u < v \\ 0 & \text{其他} \end{cases}
 \end{aligned}$$

$$\begin{aligned}
 51. \quad f_{Y_1, Y_2}(y_1, y_2) &= f_{X_1, X_2} \left(e^{-\frac{y_1^2 + y_2^2}{2}}, \arccos \frac{y_1}{\sqrt{y_1^2 + y_2^2}} \right) |J_1(y_1, y_2)| \\
 &\quad + f_{X_1, X_2} \left(e^{-\frac{y_1^2 + y_2^2}{2}}, 2\pi - \arccos \frac{y_1}{\sqrt{y_1^2 + y_2^2}} \right) |J_1(y_1, y_2)| \\
 &= \frac{1}{2\pi} e^{-\frac{y_1^2 + y_2^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{y_1^2}{2}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{y_2^2}{2}} \\
 \therefore Y_1, Y_2 \text{ 独立, 且满足 } N(0, 1) \text{ 分布}
 \end{aligned}$$

$$53. \quad F_{Z_n}(x) = P(Z_n \leq x) = P(\max(X_1, X_2, \dots, X_n) \geq 1 - \frac{x}{n})$$

$$x \leq 0 \text{ 时, } F_n(x) = 0$$

$$\begin{aligned}
 x > 0 \text{ 时, 上式} &= 1 - (P(X_1) < 1 - \frac{x}{n})^n \\
 &= 1 - (1 - \frac{x}{n})^n
 \end{aligned}$$

$$\text{当 } n \rightarrow \infty \text{ 时, } F_{Z_n}(x) \rightarrow \begin{cases} 1 - e^{-2x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

$$1. (a) \varphi(\theta) = \int_a^b \frac{e^{i\theta x}}{b-a} dx = \int_a^b \frac{\cos \theta x}{b-a} dx + i \int_a^b \frac{\sin \theta x}{b-a} dx$$

$$= \frac{\sin \theta b - \sin \theta a}{b-a} + i \frac{\cos \theta a - \cos \theta b}{b-a}$$

$$= \frac{e^{i\theta b} - e^{i\theta a}}{i\theta(b-a)}$$

$\therefore (-a, a)$ 上的平均值为

$$\varphi(\theta) = \frac{\sin a\theta}{a\theta}$$

$$(b) I(\theta) = \frac{1}{a\pi} \int_{-\infty}^{+\infty} \frac{e^{i\theta x}}{x^2+1} dx$$

$$= \frac{2}{a\pi} \int_0^{+\infty} \frac{\cos \theta x}{x^2+1} dx$$

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$$I'(\theta) = \frac{2}{a\pi} \int_0^{+\infty} \frac{-x \sin \theta x}{x^2+1} dx$$

$$= -\frac{2}{a\pi} \int_0^{+\infty} \frac{\sin \theta x}{x} dx + \frac{2}{a\pi} \int_0^{+\infty} \frac{\sin \theta x}{x(x^2+1)} dx$$

$$= -\frac{1}{a} + \frac{2}{a\pi} \int_0^{+\infty} \frac{\sin \theta x}{x(x^2+1)} dx$$

$$I''(\theta) = \frac{2}{a\pi} \int_0^{+\infty} \frac{-\cos \theta x}{x^2+1} dx = -I(\theta)$$

$$\therefore I(\omega) = C_1 e^{i\omega} + C_2 e^{-i\omega} \quad I(0) = \frac{1}{a} \quad I'(0) = -\frac{1}{a}$$

$$\therefore I(\omega) = \frac{1}{a} e^{-\omega} \quad \omega > 0$$

$$\therefore I(\omega) = \frac{1}{a} e^{-|\omega|}$$

$$\therefore \varphi_x(\theta) = e^{i\theta x} - a|\theta|$$

$$(4) \varphi(\omega) = E e^{i\omega X} = \int_{-\infty}^{+\infty} e^{i\omega x} \frac{\beta^\alpha x^{\alpha-1}}{\Gamma(\alpha)} e^{-\lambda x} dx$$

$$= \int_0^{+\infty} \frac{\beta^\alpha x^{\alpha-1}}{\Gamma(\alpha)} e^{-(\beta-i\omega)x} dx$$

$$= \int_0^{+\infty} \frac{1}{\Gamma(\alpha)} \left(1 - \frac{i\omega}{\beta}\right)^{-\alpha} \int_0^{+\infty} y^{\alpha-1} e^{-y} dy \quad (y = (\beta-i\omega)x)$$

$$= \left(1 - \frac{i\omega}{\beta}\right)^{-\alpha}$$

6.

证 $\varphi(\omega)$ 为 X 的特征函数.

* 取 $Y \sim U(-a, a)$ (2) $\varphi_Y(\omega) = \frac{\sin a\omega}{a\omega}$

$\therefore \varphi_a(\omega)$ 为 $X+Y$ 的特征函数

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$$(1) \varphi_X(\omega) = \int_0^{\infty} e^{i\omega x} e^{-x} dx = \int_0^{\infty} \cos \omega x e^{-x} dx + i \int_0^{\infty} \sin \omega x e^{-x} dx$$

$$= \frac{1}{1-i\omega} \quad (\neq) \quad \frac{1}{1+i\omega}$$

$$(2) \varphi_{X-Y}(\omega) = \varphi_X(\omega) \cdot \varphi_Y(-\omega) = \frac{1}{1+\omega^2}$$

$$(6) \underline{f_{X,Y}(x,y)} = f_{X-Y}(x) = \int_{-\infty}^{+\infty} f_X(u) f_Y(u-x) du = \frac{1}{2} e^{-|x|}$$

$$\varphi_{X-Y}(\omega) = \int_{-\infty}^{+\infty} e^{i\omega x} \cdot \frac{1}{2} e^{-|x|} dx = \frac{1}{1+\omega^2}$$

$$\underline{f_{X,Y}(x,y)} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-i\omega x} \varphi_{X-Y}(\omega) d\omega = \frac{1}{2} e^{-|x|}$$

10. 设 $X_1 \sim \text{Cauchy}(m, a)$

$$\text{则 } \varphi_{X_1}(\theta) = e^{im\theta - a|\theta|}$$

$$\text{则 } \varphi_{\frac{X_1 + X_2 + \dots + X_n}{n}} = \left(e^{im\theta - a|\theta|} \right)^n = e^{imn\theta - na|\theta|}$$

$$\therefore \frac{X_1 + X_2 + \dots + X_n}{n} \sim \text{Cauchy}(m, a)$$